SULIT

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II , 2014 / 2015

SUBJECT CODE	:	SCSI 2143
SUBJECT NAME	:	PROBABILITY & STATISTICAL DATA ANALYSIS
SECTION	:	ALL
TIME	:	9.00 AM – 12.00 NOON
DATE/DAY	:	17 / 6 / 2015 (WEDNESDAY)
VENUES	:	DEWAN SERBAGUNA, KOLEJ TUN RAZAK

INSTRUCTIONS :

Please answer ALL question in the answer booklet.

(Please Write Your Lecture Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of ELEVEN (11) printed pages including this page.

QUESTION 1 (13 MARKS)

(a) A manager of new internet provider company claims that the transmission speeds of their system is 400 kilobits per second.

(i) What are the null and alternative hypotheses from the customer point of view who want to investigate if the connection is slower than the company advertised?

(2 Marks)

(ii) What constitutes a Type I error and a Type II error for this problem? (3 Marks)

(b) A soft-drink dispensing machine is said to be out of control if the variance of the contents exceeds 1.15 deciliters. If a random sample of 25 drinks from this machine has a variance of 2.03 deciliters, does this indicate at the 0.05 level of significance that the machine is out of control? Assume that the contents are approximately normally distributed. (8 Marks)

QUESTION 2 (22 MARKS)

(a) A study was conducted to determine if the mathematic course is better understood when a tutorial session constitutes part of the course. Students were randomly selected to participate in either 3 hour lecture without tutorials or 4 hour lecture with tutorials. In the section with tutorials, 11 students made an average grade of 85 with standard deviation of 4.7, and in the section without tutorials, 17 students made an average grade of 79 with standard deviation of 6.1. Is there evidence that the tutorial session increase the average mark by as much as 8 points at the 0.05 level of significance? Assume the populations to be approximately normally distributed with equal variances. (10 Marks)

(b) **Table 1** shows the data for the third and fourth round scores of a random sample of five competitors in an open golf tournament.

Table 1

Competitor	**A**	**B**	**C**	**D**	**E**
3rd round	76	75	72	75	79
4th round	70	73	71	68	76

(i) Find the mean and standard deviation of the differences for the scores. (7 Marks)

(ii) Compute the test statistic and test whether there is a difference in the mean scores of all competitors in the two rounds at 0.05 significant level. (5 Marks)

QUESTION 3 (25 MARKS)

(a) Each observation in a random sample of 70 motorcycle accidents resulting in death was classified according to the day of the week on which the accident occurred. The data are given in **Table 2.** Based on these data, is it reasonable to conclude that the proportion of accidents is not the same for all days of the week? Use $\alpha = 0.05$. (9 Marks)

Table 2

Day of Week	Frequency
Sunday	16
Monday	11
Tuesday	8
Wednesday	7
Thursday	6
Friday	9
Saturday	13

(b) According to a simple genetic principle, if both the mother and the father of a child have genotype ***Aa***, then there is probability 1/4 that the child will have genotype ***AA***; there is probability 1/2 that he or she will have genotype ***Aa***; and there is probability 1/4 that he or she will have genotype ***aa***. In a random sample of 48 children having both parents with genotype ***Aa***, it is found that 15 have genotype ***AA***, 20 have genotype ***Aa***, and 13 have genotype ***aa***. Investigate whether the simple genetic principle is correct by carrying out a χ^2 test of goodness-of-fit. Use $\alpha = 0.01$. (5 Marks)

(c) Suppose that 150 persons are selected at random from a large population; and that each person in the sample is classified according to whether his blood type is *O*, *A*, *B*, or *AB*, and also according to whether his blood type is *Rh* positive or *Rh* negative. The observed

numbers are given in **Table 3**. Test the hypothesis that the two classifications of blood type are independent. Use $\alpha = 0.1$. (11 Marks)

Table 3

Blood Type	***O***	***A***	***B***	***AB***
Rh positive	40	44	26	10
Rh negative	6	14	4	6

QUESTION 4 (30 MARKS)

(a) **Table 4** shows the costs (in Ringgit Malaysia) of air fares for seven (7) different airlines (A, B, C, D, E, F and G) from City I to City II. The costs are based on the condition tickets purchased 30 days in advance and one day in advance.

Table 4

Airlines / Condition	**A**	**B**	**C**	**D**	**E**	**F**	**G**
30 Days (x)	244	260	264	264	278	318	280
One Day (y)	456	614	567	943	628	1088	536

(i) Which correlation coefficient will you use for the data presented in Table 4 and why? (2 Marks)

(ii) Calculate the correlation coefficient, r. (13 Marks)

(iii) Does the value of r suggest a strong linear relationship? (1 Mark)

(iv) Is there sufficient evidence to conclude that there is a linear correlation between costs of tickets purchased 30 days in advance and those purchased one day in advance? (1 Mark)

(b) All of us are confronted with the problem of determining how much of a tip we should leave a waiter or waitress in a restaurant. Unofficially, many people say that the tip should be 15% of the bill. While some of us are trying to walk a fine line between giving away too much money and looking too cheap, others take genuine pleasure in rewarding

service with a very generous tip. Consider **Table 5** which lists the data collected from a sample.

Table 5

Bill (RM), x	33.46	50.68	87.92	98.84	63.60	107.34
Tip (RM), y	5.50	5.00	8.08	17.00	12.00	16.00

(i) Fit the model $Y = \beta_0 + \beta_1 x + \varepsilon$ to these data using the method of least squares. (11 Marks)

(ii) What is your estimation for the expected change in tip value for a one-unit change in bill value? (1 Mark)

(iii) If the bill value is x = RM150.50, what would you estimate the amount of tip that should be given? (1 Mark)

QUESTION 5 (10 MARKS)

The values in the **Table 6** are measured maximum breadths (in millimetres) of male Egyptian skulls from different epochs (period of times in history). Changes in head shape over time suggest that interbreeding occurred with immigrant populations.

Table 6

4000 BC	131	138	125	129	132	135	132	134	138
1850 BC	129	134	136	137	137	129	136	138	134
150 AD	128	138	136	139	141	142	137	145	137

(i) Find the variance between samples. (2 Marks)

(ii) Estimate the variance within samples. (2 Marks)

(iii) Calculate the F test statistic. (2 Marks)

(iv) Find the numerator and denominator degrees of freedom. (1 Mark)

(v) Use a 0.05 significance level to test the claim that the mean is the same for the different epochs. (3 Marks)

List of Formula:

(a) Hypothesis Testing for one sample

$$Test\ Statistic, z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$Test\ Statistic, T_0 = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$Test\ Statistic, \chi_0^2 = \frac{(n-1)s^2}{\sigma^2}$$

(b) Hypothesis Testing for two samples

$$Test\ Statistic, z_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$Test\ Statistic, T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}; S_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$$

$$Test\ Statistic, T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}; v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

$$Test\ Statistic, F_0 = \frac{s_1^2}{s_2^2}$$

(c) Chi-Square Test

$$Test\ Statistic, \chi_0^2 = \sum_i^n \frac{(o_i - e_i)^2}{e_i}$$

$$Degree\ of\ Freedom\ for\ two-way\ contigency\ table = (r-1)(c-1)$$

(d) Correlation

$$Pearson\ Correlation\ Coefficient, r = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)\left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

$$Spearman\ Correlation\ Coefficient, r_S = 1 - \frac{6\sum d_i^2}{n(n^2-1)}\ where\ d_i = y_i - x_i$$

$$Test\ Statistic, t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

(e) Regression

$$b_1 = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

(f) Analysis of variance (ANOVA)

$$F = \frac{n s_{\bar{x}}^2}{s_p^2}$$

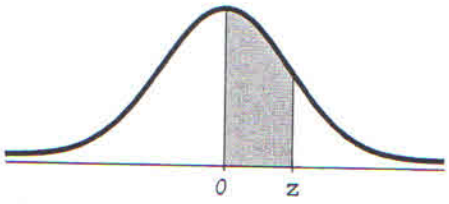


TABLE A-2 Standard Normal (z) Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495 *	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949 *	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.10 and higher	.4999									

NOTE: For values of z above 3.09, use 0.4999 for the area.
*Use these common values that result from interpolation:

z score	Area
1.645	0.4500
2.575	0.4950

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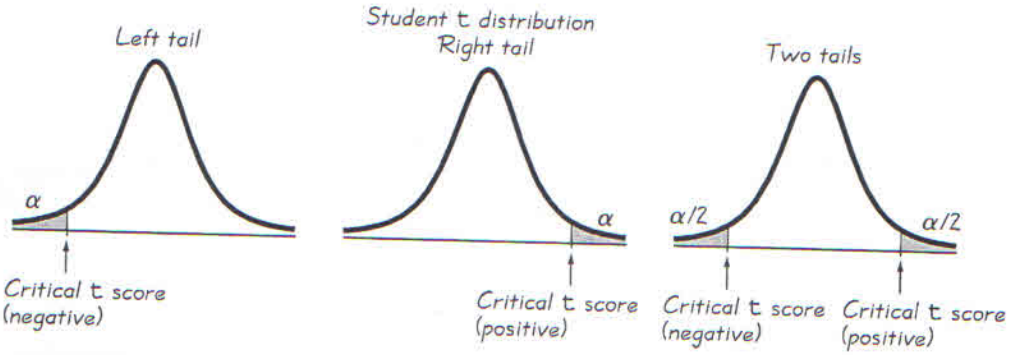


TABLE A-3 t Distribution

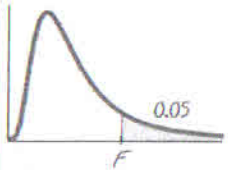
Degrees of Freedom	α: .005 (one tail) .01 (two tails)	.01 (one tail) .02 (two tails)	.025 (one tail) .05 (two tails)	.05 (one tail) .10 (two tails)	.10 (one tail) .20 (two tails)	.25 (one tail) .50 (two tails)
1	63.657	31.821	12.706	6.314	3.078	1.000
2	9.925	6.965	4.303	2.920	1.886	.816
3	5.841	4.541	3.182	2.353	1.638	.765
4	4.604	3.747	2.776	2.132	1.533	.741
5	4.032	3.365	2.571	2.015	1.476	.727
6	3.707	3.143	2.447	1.943	1.440	.718
7	3.500	2.998	2.365	1.895	1.415	.711
8	3.355	2.896	2.306	1.860	1.397	.706
9	3.250	2.821	2.262	1.833	1.383	.703
10	3.169	2.764	2.228	1.812	1.372	.700
11	3.106	2.718	2.201	1.796	1.363	.697
12	3.054	2.681	2.179	1.782	1.356	.696
13	3.012	2.650	2.160	1.771	1.350	.694
14	2.977	2.625	2.145	1.761	1.345	.692
15	2.947	2.602	2.132	1.753	1.341	.691
16	2.921	2.584	2.120	1.746	1.337	.690
17	2.898	2.567	2.110	1.740	1.333	.689
18	2.878	2.552	2.101	1.734	1.330	.688
19	2.861	2.540	2.093	1.729	1.328	.688
20	2.845	2.528	2.086	1.725	1.325	.687
21	2.831	2.518	2.080	1.721	1.323	.686
22	2.819	2.508	2.074	1.717	1.321	.686
23	2.807	2.500	2.069	1.714	1.320	.685
24	2.797	2.492	2.064	1.711	1.318	.685
25	2.787	2.485	2.060	1.708	1.316	.684
26	2.779	2.479	2.056	1.706	1.315	.684
27	2.771	2.473	2.052	1.703	1.314	.684
28	2.763	2.467	2.048	1.701	1.313	.683
29	2.756	2.462	2.045	1.699	1.311	.683
Large (z)	2.575	2.326	1.960	1.645	1.282	.675

TABLE A-4 Chi-Square (χ^2) Distribution

Degrees of Freedom	Area to the Right of the Critical Value									
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.299
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

From Donald B. Owen, *Handbook of Statistical Tables*, ©1962 Addison-Wesley Publishing Co., Reading, MA. Reprinted with permission of the publisher.

TABLE A-5 *F* Distribution ($\alpha = 0.05$ in the right tail)



Denominator degrees of freedom (df_2)	Numerator degrees of freedom (df_1)								
	1	2	3	4	5	6	7	8	9
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54
2	18.513	19.000	19.164	19.247	19.296	19.330	19.353	19.371	19.385
3	10.128	9.5521	9.2766	9.1172	9.0135	8.9406	8.8867	8.8452	8.8123
4	7.7086	6.9443	6.5914	6.3882	6.2561	6.1631	6.0942	6.0410	6.9988
5	6.6079	5.7861	5.4095	5.1922	5.0503	4.9503	4.8759	4.8183	4.7725
6	5.9874	5.1433	4.7571	4.5337	4.3874	4.2839	4.2067	4.1468	4.0990
7	5.5914	4.7374	4.3468	4.1203	3.9715	3.8660	3.7870	3.7257	3.6767
8	5.3177	4.4590	4.0662	3.8379	3.6875	3.5806	3.5005	3.4381	3.3881
9	5.1174	4.2565	3.8625	3.6331	3.4817	3.3738	3.2927	3.2296	3.1789
10	4.9646	4.1028	3.7083	3.4780	3.3258	3.2172	3.1355	3.0717	3.0204
11	4.8443	3.9823	3.5874	3.3567	3.2039	3.0946	3.0123	2.9480	2.8962
12	4.7472	3.8853	3.4903	3.2592	3.1059	2.9961	2.9134	2.8486	2.7964
13	4.6672	3.8056	3.4105	3.1791	3.0254	2.9153	2.8321	2.7669	2.7144
14	4.6001	3.7389	3.3439	3.1122	2.9582	2.8477	2.7642	2.6987	2.6458
15	4.5431	3.6823	3.2874	3.0556	2.9013	2.7905	2.7066	2.6408	2.5876
16	4.4940	3.6337	3.2389	3.0069	2.8524	2.7413	2.6572	2.5911	2.5377
17	4.4513	3.5915	3.1968	2.9647	2.8100	2.6987	2.6143	2.5480	2.4943
18	4.4139	3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563
19	4.3807	3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227
20	4.3512	3.4928	3.0984	2.8661	2.7109	2.5990	2.5140	2.4471	2.3928
21	4.3248	3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.3660
22	4.3009	3.4434	3.0491	2.8167	2.6613	2.5491	2.4638	2.3965	2.3419
23	4.2793	3.4221	3.0280	2.7955	2.6400	2.5277	2.4422	2.3748	2.3201
24	4.2597	3.4028	3.0088	2.7763	2.6207	2.5082	2.4226	2.3551	2.3002
25	4.2417	3.3852	2.9912	2.7587	2.6030	2.4904	2.4047	2.3371	2.2821
26	4.2252	3.3690	2.9752	2.7426	2.5868	2.4741	2.3883	2.3205	2.2655
27	4.2100	3.3541	2.9604	2.7278	2.5719	2.4591	2.3732	2.3053	2.2501
28	4.1960	3.3404	2.9467	2.7141	2.5581	2.4453	2.3593	2.2913	2.2360
29	4.1830	3.3277	2.9340	2.7014	2.5454	2.4324	2.3463	2.2783	2.2229
30	4.1709	3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107
40	4.0847	3.2317	2.8387	2.6060	2.4495	2.3359	2.2490	2.1802	2.1240
60	4.0012	3.1504	2.7581	2.5252	2.3683	2.2541	2.1665	2.0970	2.0401
120	3.9201	3.0718	2.6802	2.4472	2.2899	2.1750	2.0868	2.0164	1.9588
∞	3.8415	2.9957	2.6049	2.3719	2.2141	2.0986	2.0096	1.9384	1.8799